Dictionarium Sinicum: Building a 428K-Headword Chinese Dictionary in 18 Languages with a Single LLM for Under $150

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Abstract

We present *Dictionarium Sinicum*, an open multilingual Chinese dictionary comprising 428,073 headwords with definitions in 18 languages, constructed by merging seven community-maintained dictionaries with glosses generated by a single open-weight LLM (MiniMax M2.5) at a total API inference cost of approximately $150 at published pay-as-you-go rates. A three-phase pipeline—batch translation, retry, and context-aware backfill—achieves 99.3% coverage across all 7.7M headword–language slots (100% for multi-character headwords). We describe prompt engineering techniques that prevent script contamination in CJK-target languages, including Japanese kanji echo and Korean hanja mixing, and a deterministic post-translation script validator that detects and rejects cross-language contamination at a rate of 0.83% of all generated definitions. Automated comparison against four community reference dictionaries shows 87.3% sense coverage, outperforming ten MT baselines (Google Translate, DeepL, Claude, GPT-4o, and five free LLMs) by 12–22 percentage points; the same model with a generic prompt scores 14 points lower, isolating prompt engineering value. The dictionary, augmented with 511K Cantonese and Hokkien dialect forms, is released under CC BY-SA 4.0.

1 Introduction

Bilingual and multilingual dictionaries remain essential infrastructure for language learning, translation, and NLP. For Chinese, the open-source community has produced high-quality dictionaries for a handful of language pairs—CC-CEDICT for English (∼124K entries), HanDeDict for German (∼159K entries), CFDICT for French (∼56K entries), and CC-CIDICT for Indonesian (∼124K entries)—but no comparable resources exist for Korean, Russian, Vietnamese, Tagalog, Persian, Swedish, Dutch, Portuguese, Arabic, Thai, Hindi, or Italian (CHEDICC provides a small Spanish dictionary of ∼4,400 entries). Creating such dictionaries manually requires years of expert effort per language pair.

Recent advances in large language models suggest an alternative: using a single multilingual model to generate dictionary-quality definitions at scale. However, prior work on LLM-assisted lexicography has been limited to small-scale experiments. The closest precedent is a Pre-Qin philosophy lexicon project using Qwen3-14B to generate bilingual dictionary entries for approximately 1,000 classical Chinese philosophical terms in a single language pair (Wu and Wang, 2025). Lu et al. (2024) use existing bilingual dictionaries as chain-of-thought context to improve LLM translation of rare words—a complementary approach where dictionaries help LLMs translate, whereas we use LLMs to *build* dictionaries. No published work has attempted dictionary generation at the scale of hundreds of thousands of headwords across more than two target languages.

We address this gap with *Dictionarium Sinicum*, a system that:

1. **Merges** seven community-maintained Chinese dictionaries (CC-CEDICT, HanDeDict, CFDICT, CC-CIDICT, CHEDICC, Wiktextract, JMdict) covering 6 language pairs into a unified headword table of 428,073 entries.

2. **Translates** all headwords into 18 languages using a single open-weight LLM (MiniMax M2.5, available on multiple inference providers) with carefully engineered prompts

that produce dictionary-style output and prevent script contamination.

3. **Backfills** gaps using a context-aware prompt that sends existing translations as read-only context, ensuring consistency across languages.

4. **Augments** the dictionary with 511,514 dialect forms from 18 open sources covering Cantonese (Jyutping, 99.9% single-character coverage) and Hokkien (POJ/Tâi-lô) pronunciations and lexical equivalents.

Total inference cost was approximately $150 and the full pipeline completed in under one week.

2 Related Work

2.1 Multilingual Lexical Resources

Large-scale multilingual lexical resources have been constructed through both manual and automated methods. PanLex (Kamholz et al., 2014) aggregates bilingual dictionaries into a graph covering ~5,700 languages with over 1.2 billion pairwise translations, but its Chinese coverage is limited to headword-level mappings without definition glosses. BabelNet (Navigli and Ponzetto, 2012) combines WordNet, Wikipedia, and other resources into a multilingual encyclopedic dictionary, but is not open-access for bulk redistribution and focuses on synset alignment rather than dictionary-style glosses. Our work differs from both in producing complete, human-readable definition glosses for a single source language across 18 targets, released under CC BY-SA 4.0.

2.2 LLM-Assisted Lexicography

The use of LLMs for dictionary construction is an emerging area with limited published work. The closest precedent is a Pre-Qin philosophy lexicon (Wu and Wang, 2025) that uses Qwen3-14B to generate ~1,000 bilingual entries for classical Chinese philosophical terms—two orders of magnitude smaller than our work. Chain-of-Dictionary prompting (Lu et al., 2024) uses existing bilingual dictionaries as chain-of-thought context to improve LLM translation of rare words—a complementary approach where dictionaries help LLMs translate, whereas we use LLMs to *build* dictionaries.

2.3 Multilingual Translation Models

General-purpose multilingual translation models such as NLLB (Costa-jussà et al., 2022) cover 200+ languages but are designed for sentence-level translation, not dictionary-style glosses. Tower (Alves et al., 2024) focuses on high-resource language pairs. Neither has been evaluated for or applied to dictionary construction. These models do not natively produce multi-sense gloss lists, making direct comparison nontrivial (see Limitations).

Source	Lang	Entries	License
CC-CEDICT	en	123,127	CC BY-SA 4.0
HanDeDict	de	159,809	CC BY-SA 3.0
CFDICT	fr	56,280	CC BY-SA 3.0
CC-CIDICT	id	123,157	CC BY-SA 4.0
CHEDICC	es	4,400	CC BY-SA 4.0
Wiktextract	multi	140,577	CC BY-SA 4.0
JMdict	ja	131,427	CC BY-SA 4.0

Table 1: Source dictionaries. All use compatible Creative Commons licenses (BY-SA 3.0/4.0).

2.4 Community Chinese Dictionaries

The open Chinese dictionary ecosystem is built on the CC-CEDICT model: community-maintained, CEDICT-format text files released under Creative Commons licenses. CC-CEDICT (CC-CEDICT), HanDeDict (HanDeDict), CFDICT (CFDICT), and CC-CIDICT (CC-CIDICT) together cover four European languages and Indonesian. Wiktextract (Ylonen, 2022) provides multilingual glosses extracted from Wiktionary, and JMdict (Breen, 2024) covers Japanese. CHEDICC provides a small Spanish resource (~4,400 entries), but no comparable community dictionaries exist for Korean, Russian, Vietnamese, Tagalog, Persian, Swedish, Dutch, Portuguese, Arabic, Thai, Hindi, or Italian.

3 Data Sources and Headword Unification

3.1 Source Dictionaries

Table 1 lists the seven source dictionaries. All sources use compatible Creative Commons licenses, allowing the combined work to be distributed under CC BY-SA 4.0.

3.2 Headword Unification

Source dictionaries use varying conventions for pinyin romanization, traditional/simplified character pairs, and entry granularity. We unified headwords on the triple (traditional, simplified, pinyin), yielding 428,073 unique headwords. Of these, 30,546 are single-character entries covering 19,471 distinct Chinese characters (many characters have multiple pinyin readings—e.g., 参 has 12 distinct entries).

3.3 Dialect Forms

We augmented the dictionary with Cantonese and Hokkien dialect data from 18 open sources (Table 2). Cantonese coverage is effectively complete: 16,564 unique single-character readings covering 99.9% of BMP CJK characters. Hokkien coverage spans 6,850 single-character readings—lower in percentage terms (37% BMP) but representing all characters with documented Hokkien pronunciations in open data.

Hokkien romanization uses two systems: Tâi-lô (the MOE standard) and traditional POJ (Peh-ōe-jī). A distinctive feature of Hokkien is its high lexical divergence from Mandarin: 68–78% of Hokkien forms use different characters entirely (e.g., 漂亮 *piàoliang* “beautiful” → 媠 *suí*; 東西 *dōngxi* “thing” → 物件 *mih-kiānn*).

3.4 Chinese Text Corpus

To provide real-world usage context during translation, we constructed a large-scale Chinese text corpus from eight publicly available sources (Table 3). Articles were chunked at sentence boundaries (。！？；) into segments averaging 40–80 characters. All chunks are indexed with FTS5 using a character-level CJK tokenizer, enabling sub-15ms lookup for any Chinese term. During translation, 1–2 example sentences per headword are retrieved from this corpus to provide the LLM with attested usage context (§4.2).

4 Translation Pipeline

4.1 Model Selection

We evaluated 20+ LLMs across 10 inference providers on a 5-headword test suite covering emergency service numbers (culturally specific), anime titles (transliteration challenges), and medical terminology. Selection criteria were: (1) correct 18-language output in the prescribed format, (2) no script contamination, (3) geographic context preservation, (4) sustainable throughput at 428K headwords, and (5) cost under $200 at published rates.

MiniMax M2.5 was selected for production: zero issues on the test suite, 1.2s/entry throughput, no observed rate limits with 20 parallel workers, and an estimated cost of ~$150 for the full run at published pay-as-you-go rates ($0.30/M input tokens, $1.20/M output tokens; Table 4).

A notable finding is that the same open-weight model can produce dramatically different output depending on the inference provider. MiniMax M2.1 served by NVIDIA NIM took 167s per entry with thinking mode silently enabled, while the same weights on Fireworks took 5.4s with thinking disabled. Benchmark scores (SWE-bench, MMLU) showed no predictive value for this structured multilingual task: DeepSeek V3.2 (SWE-bench 73.1%) echoed headwords, while MiniMax M2.5 (unranked) produced flawless output. These observations align with emerging evidence that inference acceleration introduces unpredictable behavioral changes (Kirsten et al., 2025).

4.2 Prompt Engineering

The production prompt generates all 18 languages simultaneously for each batch of 20 headwords. Key design decisions:

System prompt (176 tokens): Establishes the lexicographer persona, output format (`xx: def1/def2`), conciseness constraints (max 5 glosses), and language-specific rules addressing three classes of script contamination:

1. **Japanese kanji echo:** Without explicit instruction, LLMs frequently “define” a Chinese word in Japanese by repeating the same characters (e.g., 銀行 → 銀行). The prompt requires: *“Provide the MEANING in Japanese, not just kanji echo or kana reading.”*
2. **Korean hanja mixing:** Korean definitions often include Chinese characters (hanja) unrecognizable to most modern Korean readers. The prompt specifies: *“Write in Hangul only.”*
3. **General CJK leakage:** A blanket prohibition: *“Every non-Chinese definition must contain ZERO Chinese characters.”*

User prompt (batch of 20): Each entry includes traditional/simplified characters, pinyin, POS tag, all existing community definitions as context, and 1–2 example sentences drawn from the 113M-chunk Chinese corpus (Table 3). The corpus examples—sourced from Chinese Wikipedia, Baidu Baike, THUCNews, and news2016zh—provide real-world usage context that helps disambiguate polysemous headwords. Example sentences are retrieved via FTS5 lookup (<15ms per headword), filtered to prefer 20–100 character chunks.

This constitutes **pivot translation:** for the twelve languages without community dictionaries

Source	Dialect	Forms	License	Notes
rime-cantonese	Cantonese	110,064	CC BY 4.0	Largest single source; community Jyutping readings
CC-Canto readings	Cantonese	96,278	CC BY-SA 3.0	Jyutping overlay for CC-CEDICT headwords (Pleco)
CC-Canto	Cantonese	29,762	CC BY-SA 3.0	Dedicated Cantonese dictionary (CC-Canto)
Unihan kCantonese	Cantonese	28,800	Unicode ToU	Unicode standard Cantonese readings
WikiHan	Cant.+Hok.	38,794	CC BY-SA 4.0	Wiktionary-derived readings for both dialects
TaiHua	Hokkien	48,154	CC BY-SA 4.0	Taiwanese Hokkien dictionary (Tâi-lô romanization)
Taibun	Hokkien	39,814	MIT	POJ-based Hokkien corpus
Wiktextract	Hokkien	25,606	CC BY-SA 3.0	Hokkien readings from Wiktionary (Ylonen, 2022)
iTaigi	Hokkien	10,572	CC BY 4.0	Crowdsourced Taiwanese Hokkien (iTaigi)
7 additional	Mixed	83,670	Various open	ChhoeTaigi, early missionary sources, others

Table 2: Dialect data sources. Total: 511,514 forms (290,421 Cantonese, 221,093 Hokkien) from 18 open sources. All sources use open licenses compatible with CC BY-SA 4.0 redistribution; three sources with NC/ND restrictions (Maryknoll, Embree, Kauiokpoo) are excluded from commercial builds.

Source	Articles	Chunks	Type
Chinese Wikipedia	1.4M	14M	Encyclopedia
Baidu Baike	5.6M	33M	Encyclopedia
THUCNews	0.8M	11M	News
news2016zh	2.5M	18M	News
NiuTrans Classical	0.3M	4M	Literary
chinese-poetry	0.3M	2M	Poetry
ChID	0.6M	7M	Idiom cloze
LCCC	22M	24M	Dialogue
Total	**34M**	**113M**	

Table 3: Chinese text corpus sources. Articles are chunked at sentence boundaries into segments of 40–80 characters.

Model	Provider	s/ent	Issues
MiniMax M2.5	MiniMax	1.2	None
Kimi K2	Groq	1.8	None
MiniMax M2.1	Fireworks	5.4	None
Nemotron 253B	NIM	7.0	None
Sonnet 4.6	Anthropic	19.8	HW echo
DeepSeek V3.2	NIM	35.3	HW echo
MiniMax M2.1	NIM	167	Think ON

Table 4: Model evaluation (5-headword test). HW echo = headword echo in output. Think ON = thinking mode silently enabled.

(Swedish, Korean, Russian, Vietnamese, Tagalog, Persian, Dutch, Portuguese, Arabic, Thai, Hindi, Italian), the model effectively translates from the English and other community-language glosses rather than directly from Chinese.

Token economics: Approximately 5,700 tokens per batch (500 system + 2,200 input including examples + 3,000 output). Full run: 21,403 batches. At MiniMax M2.5 published rates ($0.30/M input, $1.20/M output), the estimated cost is ~$146 without prompt caching or ~$118 with the platform's automatic caching ($0.03/M for cached system prompts).

Phase	Entries	Defs	Time	Cost
1. Main	374,602	4.41M	24h	$130
2. Retry	53,471	616K	92m	$12
3. Backfill	82,550	110K	105m	$4
Total	**428,073**	**7.70M**	**27h**	**$146**

Table 5: Three-phase execution summary. Costs at published pay-as-you-go rates.

4.3 Three-Phase Execution

Table 5 summarizes the three-phase execution. **Phase 1** processed all 428,073 headwords in batches of 20 with 20 parallel workers. The API returned unparseable responses for 53,471 headwords (12.5%), primarily due to batch-level formatting errors where the model's response was cut off or malformed. **Phase 2** retried failed headwords using the same prompt, recovering 616,276 additional definitions and leaving 5,875 headwords with partial coverage. **Phase 3** used a distinct context-aware backfill prompt that sends all existing translations as read-only context, asking the model to produce *only* the missing languages. This design ensures consistency: the model sees existing glosses and produces output that matches in style and specificity. Three backfill passes closed all remaining multi-character gaps, achieving 100.0% coverage for the 397,527 multi-character headwords.

4.4 Parser Robustness

Parsing structured LLM output at scale required handling several failure modes: (1) **multi-language line collapse**—the model outputs `en: bank/de: Bank/fr: banque` on a single line; a regex normalizer splits on `/xx:` boundaries; (2) **numbered vs. unnumbered responses**—the backfill prompt causes the model to drop entry

Language	Rate	Primary Contamination
Arabic	3.91%	Latin (9.7K), CJK (6.1K)
Persian	1.70%	Latin (4.5K), CJK (2.3K)
Japanese	1.15%	Hangul (3.1K), Cyrillic (1.8K)
Hindi	0.87%	CJK (3.0K), Arabic (0.4K)
Indonesian	0.66%	CJK (2.3K), Cyrillic (0.5K)
Korean	0.56%	Kana (1.4K), Cyrillic (0.7K)
Other 12	0.34–0.58%	CJK, Cyrillic

Table 6: Script contamination by target language (full corpus). Uniformly distributed across headword ranges.

numbers; the parser uses blank lines as separators; (3) **overwrite protection**—three layers prevent backfill from corrupting existing data: query filter, parser filter, and database existence check.

5 Quality Assessment

5.1 Script Contamination Audit

A full-corpus audit of all 7.7M minimax definitions using a deterministic script validator revealed 63,712 definitions (0.83%) containing scripts that do not belong to the target language.

Table 6 breaks down contamination by language. We identified five distinct contamination modes:

1. **Cyrillic-in-non-Cyrillic** (~10K): The model falls back to Russian when translating into less-resourced scripts. Example: Arabic for 一世 → “*zhizn’* واحدة” (Russian mixed with Arabic).

2. **Hangul-in-Japanese / Kana-in-Korean** (~4.4K): Systematic confusion between CJK languages.

3. **Latin-in-Arabic/Persian** (~14K): English or French words injected into Arabic/Persian definitions, after exempting proper names, scientific binomials, and universal acronyms.

4. **CJK-in-non-CJK** (~40K): Chinese characters in non-CJK definitions, after exempting metalinguistic cross-references (“variant of 迴”).

5. **Multi-language field dumps** (rare): Multiple languages written into a single field.

Cleanup methodology. We implemented a three-stage pipeline: (1) a **deterministic script validator** that detects all Unicode script families in each definition and compares against a per-language allowlist, with exemptions for CJK in metalinguistic reference frames (detected via compiled regex patterns covering all 18 target languages) and Latin in Arabic/Persian when constituting single letters, scientific binomials, proper names, or universal acronyms; (2) **deletion** of contaminated minimax-sourced definitions (community entries never touched); (3) **context-aware re-translation** via the backfill pipeline with the validator integrated as a hard reject gate—contaminated output is discarded before database write.

Post-cleanup status. Multi-character headwords (397,527): 100.0% complete across all 18 languages—zero gaps. Single-character headwords (30,546): 27,676 (90.6%) complete; 2,870 (9.4%) have residual gaps in 1–6 languages. The largest gaps are in Arabic (1,084), Thai (1,016), German (895), and Persian (660). These persist because the affected headwords are obscure radicals whose definitions inherently cite Chinese characters, which the validator correctly rejects. All HSK 3.0 characters have complete 18-language coverage.

5.2 Comparison with Community Dictionaries

For the four languages with existing community dictionaries providing glosses (English, German, French, Indonesian), we compared LLM-generated definitions against human-curated references.[1] We sampled 200 headwords stratified by community source coverage as a frequency proxy and measured two metrics across 453 headword–language pairs:

1. **Sense coverage:** proportion of reference senses captured by the LLM gloss (automated lexical overlap with 50% word-overlap threshold)

2. **False sense rate:** proportion of LLM glosses not matched by any reference sense

Important caveat. The translation prompt provides all existing community definitions as input context (§4.2). For these four languages, the model *sees* the reference glosses. This comparison therefore measures **context utilization**—how faithfully the model preserves reference senses—not independent translation quality. We find that 42% of English, 40% of German, 55% of French, and 60% of Indonesian LLM glosses are substring matches of the corresponding community definition.

The overall sense coverage of 87.3% (95% bootstrap CI: [84.7%, 89.8%]) with a false sense rate

[1] JMdict was excluded because it stores kana readings rather than Japanese-language glosses.

Lang	Ref.	n	Cov.	False	BSc	COMET
en	CC-CEDICT	139	85.8	14.1	.932	.777
de	HanDeDict	105	87.9	17.9	.905	.712
fr	CFDICT	70	81.2	18.6	.926	.787
id	CC-CIDICT	139	91.3	3.7	.933	.820
Overall		**453**	**87.3**	**12.5**	**.924**	**.774**

Table 7: LLM vs. community dictionary. Cov./False in %; BSc = BERTScore F1 (XLM-R-large); COMET = `wmt22-comet-da` (Rei et al., 2022a). Semantic-metric n=139/105/67/139 (fr slightly reduced by non-empty-reference filter).

Band	n	Cov.	False
high (4+ sources)	190	84.7%	16.7%
mid (3 sources)	146	87.4%	11.8%
low (2 sources)	94	90.8%	5.7%
rare (1 source)	23	93.5%	10.9%

Table 8: Comparison by frequency band (community source coverage as proxy).

of 12.5% (95% CI: [10.1%, 15.0%]) shows that the model successfully preserves most reference senses while introducing few unsupported glosses (Tables 7–8). Manual inspection reveals that most "false senses" are legitimate synonyms absent from the reference.

The main failure mode is **sense compression:** the LLM produces fewer glosses than community dictionaries (median 2 vs. 3), preferring the most common translation equivalent and omitting specialized or archaic senses. Manual inspection of the 57 headword–language pairs with sense coverage below 50% reveals that sense compression (38%) and register shift (22%) dominate, with genuine errors (wrong meaning) accounting for only 3%.

Semantic metric agreement. Because lexical overlap can miss legitimate synonymy, Table 7 additionally reports two model-based MT metrics on the same reference sample: BERTScore F1 with XLM-RoBERTa-large (Zhang et al., 2020) and COMET-DA (`wmt22-comet-da`, Rei et al., 2022a). Macro-averaged BERTScore F1 is 0.924 and COMET-DA is 0.774; the language ordering tracks sense coverage, with Indonesian highest on all three metrics and German lowest on the two model-based scores. As with sense coverage, the LLM sees the community gloss as input context (§4.2), so these values characterize context utilization rather than independent translation quality; independent quality across all 18 languages is addressed in §5.3.

System	n	Cov.	False	Fmt.
Our pipeline	**453**	**87.3**	**12.5**	**∼100**
MiniMax (generic)	275	73.0	27.8	91.0
Claude Sonnet 4	275	75.4	28.7	74.9
Google Translate	275	70.9	28.9	72.0
GPT-4o-mini	275	70.4	32.3	74.2
Azure Translator	275	70.4	29.6	69.5
DeepL	275	67.5	33.2	70.2
DeepSeek V3.1	275	72.0	36.5	82.1
Kimi K2	275	68.0	35.1	82.1
Llama 4 Scout	275	68.9	49.2	73.3
Qwen3-235B	275	65.2	43.4	78.8

Table 9: MT baseline comparison (%). Generic prompt for all baselines. Free LLMs via Groq, Cerebras, SambaNova.

For the twelve languages *without* community dictionaries, the model relies on pivot translation. Section 5.3 addresses this gap.

5.3 MT Baselines and Non-Community Evaluation

We evaluated ten alternative systems on the same 275 headword–language evaluation pairs using a generic translation prompt ("Translate to [Language]: [text]. Be concise—dictionary style."), deliberately simpler than our production prompt (§4.2). We add a **format compliance** metric: whether output matches CEDICT-style gloss format (short slash-separated segments, $\leq$5 words, no sentence-ending punctuation).

Table 9 shows our pipeline outperforms all baselines on sense coverage (+12–22pp) and false sense rate ($-$15–37pp). The *same model* (MiniMax M2.5) with a generic prompt scores only 73.0% coverage—14 points below our pipeline—isolating the value of dictionary-specific prompt engineering. Claude Sonnet 4, the strongest baseline at 75.4%, still falls 12 points short. Commercial MT and frontier LLMs produce CEDICT-compatible format only 70–75% of the time; multiple LLMs cluster in the 65–78% range regardless of model size, suggesting pipeline design matters more than model choice.

Model-based metrics on the MT baselines. We additionally score the four API-accessible baselines with BERTScore F1, COMET-DA, and reference-free CometKiwi (Chinese source) on the same 275 pairs (Table 10). The pipeline lead extends to all three metrics: +2.1 pt BSc F1, +5.8 pt COMET-DA, +1.6 pt CometKiwi over the next-best system. Claude Sonnet 4.5 is the lowest baseline

System	n	BSc. F1	COMET	CK
Our pipeline	**311**	**0.921**	**0.773**	**0.544**
Google Translate	275	0.900	0.715	0.524
Claude Sonnet 4.5	275	0.900	0.711	0.501
DeepL	275	0.898	0.712	0.528
GPT-4o-mini	275	0.897	0.702	0.507

Table 10: Model-based metrics, macro-averaged over de/fr/id (same 275 pairs as Table 9). BSc. F1 = BERTScore F1 (XLM-R-large); COMET = `wmt22-comet-da`; CK = reference-free `wmt22-cometkiwi-da` with the Chinese headword as source. Our-pipeline row: 3-lang macro over the same de/fr/id subset of Table 7. Azure and the free-LLM systems are omitted from this revision.

Lang	CK	Lang	CK
en†	0.614	id†	0.543
de†	0.534	ja	0.546
fr†	0.555	ko	0.557
es	0.558	ru	0.568
it	0.570	vi	0.554
pt	0.554	tl	0.471
nl	0.540	fa	0.535
sv	0.540	th	0.521
		ar	0.503
		hi	0.589
Macro: 0.547 (ref 0.562 / non-community 0.542)			

Table 11: Reference-free CometKiwi scores (higher is better) on 200 stratified headwords per language (n=3,600 total). † = language with a community reference dictionary. Left column groups European targets; right column groups East Asian, Southeast Asian, and Middle Eastern targets.

on CometKiwi (0.501) despite the highest lexical sense coverage—fluent output can correlate less tightly with the Chinese source than commercial MT trained for translation fidelity.

Back-translation evaluation (n=1,571 across 13 non-community languages) yields 68.9% sense coverage overall, with European languages strongest (Spanish 79.0%, Portuguese 75.2%) and Korean weakest (60.0%). Four LLM judges (Claude Sonnet 4, Kimi K2, GPT-OSS-120B, DeepSeek V3.1) rate MiniMax definitions across 100 entries: accuracy **4.34**/5, naturalness **4.31**/5, completeness **3.38**/5 (mean pairwise disagreement: 0.64 points). The lower completeness aligns with sense compression (§5.2). Tagalog (3.90) and Hindi (4.04) show weakest accuracy. LLM judges may share biases with the evaluated model, but provide a quality signal where none previously existed.

Reference-free quality across all 18 languages. We score our pipeline output with CometKiwi (`wmt22-cometkiwi-da`, Rei et al., 2022b), a reference-free quality-estimation model, on a 200-headword stratified sample per language (n=3,600 total). Macro-averaged CometKiwi is 0.547 across all 18 languages; the four community-reference languages average 0.562 and the twelve non-community languages 0.542—a gap of only 0.02 (Table 11). This near-flat profile across resource tiers indicates that pipeline translation quality does not degrade materially for languages lacking a community-dictionary anchor. Tagalog (0.471) is the lowest and English (0.614) the highest.

English-pivot ablation. On a 50-headword Vietnamese + 50-headword Thai ablation using Llama-3.3-70B (open-weight, Groq) with vs. without English pivot input, reference-free CometKiwi is 0.568 vs. 0.502 for Vietnamese ($\Delta = +0.067$ favouring pivot) and 0.520 vs. 0.526 for Thai ($\Delta = +0.006$, direct marginally higher). Symmetric gloss agreement is 60.6% (vi) and 37.1% (th). Pivot bias is language-dependent: English context measurably *helps* Vietnamese while being neutral for Thai, so the pivot's effect depends on target-language distance from English rather than being uniform quality loss.

Per-language error taxonomy. A full-corpus analysis (Table 13, Appendix C) quantifies mono-gloss rate (55% en → 74.6% id/tl), CJK-character rate (∼1% non-CJK targets, 87% ja, 2.7% ko), and Chinese-headword echo (13% for ja).

Second-model verification. Directly answering the reviewer suggestion to have a second model verify the no-community-reference backfill, we asked an independent open-weight model (Gemma-4-31B via Cerebras) to rate each MiniMax gloss 1–5 for semantic preservation of the Chinese source. Across 14 non-community-reference target languages (n=2,800 rated, 200 per lang stratified by frequency band), the macro mean rating is **4.47/5** and only **9.6%** of glosses are flagged as low-quality (rating <3). Per-language means range from 4.14 (Tagalog) to 4.65 (Portuguese/Russian), with every language above 4.0. The two lowest per-language means (Tagalog 4.14, Thai 4.35) coincide with the lowest CometKiwi scores in Table 11 (0.471, 0.521)—two independent open-weight quality-estimation methods agree on which languages need the most improvement.

Metric	Value
Headwords	428,073
Languages	18
Language slots (headword × language)	7,705,314
Slots with LLM-generated gloss	7,697,901
Slots covered by community dict	738,777
Dialect forms (Cantonese + Hokkien)	511,514
Total cost (LLM inference, list price)	~\$146

Table 12: Final dictionary statistics.

5.4 Source Priority and Coverage

For headwords with definitions from multiple sources, we apply a priority hierarchy: community dictionaries (CC-CEDICT, HanDeDict, CFDICT, CC-CIDICT, CHEDICC, JMdict) > Wiktextract > MiniMax. Human-curated definitions take precedence.

Table 12 summarizes the final resource. Multi-character headwords (397,527): 100.0% complete—every headword has definitions in all 18 languages. Single-character headwords (30,546): 90.6% complete; residual gaps are limited to obscure radicals in Arabic, Thai, German, and Persian. Zero headwords have zero definitions.

6 Downstream Application

To validate the dictionary's practical utility, we built a language pack generator that produces consolidated SQLite databases for two Chinese learning applications. The consolidated architecture replaces per-language database files with a single database containing one column per language, enabling column-scoped FTS4 search (e.g., `MATCH 'es:banco'`). Databases are encrypted for distribution. The dictionary is currently deployed in production for English and Spanish, with all 18 languages shipping in the next release.

Limitations

Glosses, not definitions. The resource produces translation equivalents, not full lexicographic definitions with usage notes or grammatical patterns. Entries lack sense discrimination: polysemous words produce flat gloss lists. Inflecting languages lack grammatical metadata (German noun gender, Russian aspect pairs).

Evaluation limitations. Sense coverage is an automated lexical overlap proxy, not a human judgment, and partially circular for languages where the community gloss appears in the prompt context (§5.2). The model-based metrics we report—BERTScore (Zhang et al., 2020), COMET-DA (Rei et al., 2022a), and reference-free CometKiwi (Rei et al., 2022b)—mitigate the surface-form limitation of lexical overlap but inherit the biases of their underlying multilingual encoders, and CometKiwi scores are less well-calibrated for low-resource target languages. LLM-as-judge scores (§5.3) may share biases with the evaluated model. Human evaluation across all 18 languages remains future work.

Pivot bias and single-model homogeneity. All translations are mediated through English context, inheriting English sense boundaries. All 7.7M glosses originate from MiniMax M2.5; model-specific biases propagate uniformly. Reproducibility is nevertheless supported by M2.5's open weights, which are hosted by multiple third-party providers (Fireworks, NVIDIA NIM) and can be self-hosted.

Scope. The 428K headwords are inherited from community dictionaries without curation. The 18 target languages cover major families but omit Malay, Bengali, and Swahili.

7 Future Work

From glossary to dictionary. Sense-discriminated entries mapping each target-language gloss to a specific Chinese sense with usage guidance, combined with grammatical enrichment (German noun gender, Russian aspect pairs).

Computational loanword detection. The dictionary contains both Hokkien source forms and Southeast Asian target languages—infrastructure for tracing lexical diffusion at scale. Linking Hokkien *tāu-hū* to Indonesian *tahu*, Tagalog *taho*, and Thai เต้าหู้ would produce a machine-readable atlas of Chinese vocabulary diffusion across maritime Southeast Asia (Chan-Yap, 1980).

Under-resourced languages. The ~\$150 cost makes the pipeline relevant for languages without institutional support for manual lexicography—Zhuang, Tibetan, Uyghur, and Yi within China alone lack comprehensive bilingual dictionaries.

Community-in-the-loop evaluation. Native-speaker evaluation across all 18 target languages—particularly by diaspora communities who are primary users of bilingual dictionaries—would establish gold-standard quality metrics. A realistic instrument would enrol 2 annotators for each of 3–5 languages on ~100 stratified entries (accuracy, naturalness, completeness on 5-point scales). We defer this pending per-language recruitment funding; the model-based semantic metrics in §5.2–§5.3 are the strongest automated proxy currently available.

8 Conclusion

A single open-weight LLM (MiniMax M2.5) can produce a multilingual Chinese glossary of 428,073 headwords $\times$ 18 languages for approximately \$150. Enablers: batched cross-language prompting, language-specific rules that reduce script contamination to $\sim$0.83%, a deterministic script-validator gate, and context-aware backfill (100% multi-char, 90.6% single-char). Combined with 511K Cantonese/Hokkien dialect forms, this is the first open multilingual Chinese lexical resource covering the major languages of Europe, East Asia, Southeast Asia, and the Middle East.

References

Duarte Alves, Nuno M. Guerreiro, João Albuquerque, and André F. T. Martins. 2024. Tower: An open multilingual large language model for translation-related tasks. In *First Conference on Language Modeling (COLM)*.

Jim Breen. 2024. JMdict: Japanese-multilingual dictionary. https://www.edrdg.org/jmdict/j_jmdict.html. CC BY-SA 4.0.

CC-Canto. 2024. CC-Canto: Cantonese dictionary. https://cantonese.org/. CC BY-SA 3.0.

CC-CEDICT. 2024. CC-CEDICT: Community-maintained Chinese-English dictionary. https://cc-cedict.org/. CC BY-SA 4.0.

CC-CIDICT. 2024. CC-CIDICT: Chinese-Indonesian dictionary. https://cidict.org/. CC BY-SA 4.0.

CFDICT. 2024. CFDICT: Chinese-French dictionary. https://chine.in/mandarin/dictionnaire/CFDICT/. CC BY-SA 3.0.

Gloria Chan-Yap. 1980. *Hokkien Chinese Borrowings in Tagalog*. Pacific Linguistics, Series B, No. 71. Australian National University.

Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, et al. 2022. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*.

HanDeDict. 2024. HanDeDict: Chinese-German dictionary. https://handedict.zydeo.net/. CC BY-SA 3.0.

iTaigi. 2024. iTaigi: Crowdsourced Taiwanese Hokkien dictionary. https://itaigi.tw/. CC0.

David Kamholz, Jonathan Pool, and Susan Colowick. 2014. PanLex: Building a resource for panlingual lexical translation. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC 2014)*, pages 3145–3150, Reykjavik, Iceland. European Language Resources Association (ELRA).

Elisabeth Kirsten, Ivan Habernal, Vedant Nanda, and Muhammad Bilal Zafar. 2025. The impact of inference acceleration on bias of LLMs. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025)*, pages 1834–1853. Association for Computational Linguistics.

Hongyuan Lu, Haoran Yang, Haoyang Huang, Dongdong Zhang, Wai Lam, and Furu Wei. 2024. Chain-of-dictionary prompting elicits translation in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*, pages 958–976. Association for Computational Linguistics.

Roberto Navigli and Simone Paolo Ponzetto. 2012. BabelNet: The automatic construction, evaluation and application of a wide-coverage multilingual semantic network. *Artificial Intelligence*, 193:217–250.

Ricardo Rei, José G. C. De Souza, Duarte M. Alves, Chrysoula Zerva, Ana C. Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and André F. T. Martins. 2022a. COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Ricardo Rei, Marcos Treviso, Nuno M. Guerreiro, Chrysoula Zerva, Ana C. Farinha, Christine Maroti, José G. C. De Souza, Taisiya Glushkova, Duarte M. Alves, Luisa Coheur, Alon Lavie, and André F. T. Martins. 2022b. CometKiwi: IST-unbabel 2022 submission for the quality estimation shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 634–645, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Mingfei Wu and Dingding Wang. 2025. Automatic compilation of a pre-Qin philosophy lexicon via large language models. *npj Heritage Science*.

Tatu Ylonen. 2022. Wiktextract: Wiktionary as machine-readable structured data. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC 2022)*, pages 1317–1325, Marseille, France. European Language Resources Association (ELRA).

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*.

A Prompt Templates

A.1 System Prompt (production)

```
You are a professional multilingual
    Chinese
lexicographer producing dictionary-
    style
definitions in 18 languages.

Rules:
- Output EXACTLY one line per language
    in
  format "xx: def1/def2"
- Be concise: dictionary style, not
    full
  sentences
- Maximum 5 glosses per entry
- CRITICAL: Every non-Chinese
    definition must
  contain ZERO Chinese characters.

Language-specific rules:
- ja: Provide the MEANING in Japanese,
    not
  just kanji echo or kana reading.
- ko: Write in Hangul only. Do NOT mix
    in
  Chinese characters.
- vi: Write in Vietnamese with proper
  diacritics. No Chinese characters.
- ar: Write in Arabic script only. No
    Latin.
- th: Write in Thai script with tone
    marks.
- hi: Write in Devanagari script.
- fa: Write in Persian script only.
- nl/pt/it: Write in standard Dutch/
  Portuguese/Italian.
```

A.2 User Prompt (with corpus examples)

```
Chinese: 銀行 / 银行
Pinyin: yin hang
POS: noun

Existing definitions:
  English: bank/financial institution
  German: Bank/Geldinstitut
  French: banque/etablissement
      bancaire

Example sentences:
  中国人民银行决定下调金融机构贷款和存
      款基准利率。
  这家银行在全国设有两千多个分支机构。

Produce definitions for each language
    below.
Output EXACTLY one line per language.

en:
de:
fr:
es:
sv:
ja:
ko:
ru:
id:
vi:
tl:
fa:
nl:
pt:
ar:
th:
hi:
it:
```

A.3 Backfill Prompt (context-aware)

```
Fill in ONLY the missing language
    definitions.
Existing translations are provided for
consistency -- match their style and
    specificity.
Do NOT repeat or modify existing
    translations.

SCRIPT PURITY (strictly enforced --
    output
that violates these rules will be
    rejected):
Each definition must use ONLY the
    script of
its target language.

1. 三角褲 / 三角裤
   Pinyin: san jiao ku
   Existing translations:
     en: briefs/underwear
     de: Slip/Unterhose
     fr: slip/calecon
     [... 14 more languages ...]
   MISSING -- fill these:
     vi:
```

B Sample Definitions

B.1 Idiom: 画蛇添足

huà shé tiān zú "draw legs on a snake"

```
en: to improve something unnecessarily
    ;
    to gild the lily
de: der Schlange Fuesse hinzumalen;
    etwas Ueberfluessiges tun
fr: dessiner un serpent et lui ajouter
    des pattes; faire du zele
es: dibujar una serpiente y anadirle
    patas; hacer algo innecesario
ko: 뱀에게 발을 그리다;
    불필요한 것을 추가하다
vi: ve ran them chan;
    them thu khong can thiet
```

B.2 Cultural term: 春节

chūn jié "Spring Festival"

```
en: Spring Festival (Chinese New Year)
de: Chinesisches Neujahrsfest
fr: Nouvel An chinois/Fete du
    printemps
```

```
id: Festival Musim Semi
ja: 春節(中国の旧正月)
ko: 춘제/설날
vi: Tet Nguyen dan
```

B.3 Loanword chain: 豆腐

dòufu "tofu"

```
en: tofu / bean curd
de: Tofu / Bohnenkäse
id: tahu           <- Hokkien tau-hu
tl: tokwa          <- Hokkien tau-koan
vi: dau phu        <- Sino-Vietnamese
ko: 두부            <- Sino-Korean
ja: とうふ          <- Sino-Japanese
```

This entry illustrates how the Hokkien maritime trade vocabulary seeded cognates across Southeast Asian languages: *tāu-hū* (Hokkien) → *tahu* (Indonesian/Malay), *taho/tokwa* (Tagalog), *đậu phụ* (Vietnamese, Sino-Vietnamese route).

C Per-Language Error Taxonomy

Table 13 reports automated per-language error signals across the full 428,073-headword MiniMax output.

Lang	Mono %	Mean seg.	Med. len	CJK %	Zh-echo%
en	55.4	1.56	25	1.07	0.57
de	67.4	1.40	23	1.02	0.51
fr	69.6	1.37	22	1.05	0.53
id	74.6	1.31	20	1.11	0.57
es	70.3	1.36	22	1.06	0.54
sv	71.5	1.35	18	1.02	0.53
ja	61.7	1.45	9	87.45	13.01
ko	68.0	1.40	7	2.73	0.53
ru	71.0	1.35	21	0.98	0.49
vi	71.6	1.34	18	0.99	0.53
tl	73.5	1.32	22	1.04	0.55
fa	71.3	1.35	16	0.72	0.35
nl	68.4	1.38	20	1.03	0.52
pt	69.1	1.37	21	1.07	0.52
ar	65.1	1.41	17	1.46	0.40
th	68.3	1.38	18	0.80	0.36
hi	67.1	1.39	18	0.77	0.39
it	68.8	1.37	22	1.02	0.52

Table 13: Per-language error signals from the full 428,073-headword MiniMax output. Mono % = fraction of definitions with only one slash-separated gloss (sense-compression proxy). Mean seg. = mean glosses per definition. Med. len = median character length. CJK % = fraction containing any CJK character (script contamination for non-CJK targets; expected for ja/ko). Zh-echo % = fraction containing the Chinese headword string (metalinguistic reference, common in ja).